

Jyot Antani, PhD

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Expertise: Biophysics, Microbiology, Virology, Host-pathogen Interactions, Microscopy

Experience & Education

Jun 2021- Now	Postdoctoral Research Associate, Yale University (US) Paul Turner Lab , Department of Ecology and Evolutionary Biology Thierry Emonet Lab , Molecular, Cellular, and Developmental Biology Research: Viral target search on host surfaces - Dynamics, Evolution, and Therapeutic Application
Aug 2016- May 2021	PhD, Pushkar Lele Lab , Department of Chemical Engineering, Texas A&M University (US) Dissertation title: Sensory Functions of the Bacterial Flagellar Motor
Jul 2012- May 2016	Bachelor of Technology with Honours in Chemical Engineering Indian Institute of Technology Bombay (IIT Bombay), Mumbai (India)

Published Work

[Google Scholar Link](#)

- [Antani JD](#), Gupta R, Lee AH, Rhee KY, Manson MD, Lele PP (2021) Mechanosensitive recruitment of stator units promotes binding of the response regulator CheY-P to the flagellar motor. *Nature Communications*. doi: [10.1038/s41467-021-25774-2](https://doi.org/10.1038/s41467-021-25774-2)
- [Antani JD](#), Sumali AX, Lele TP, Lele PP (2021) Asymmetric random walks reveal that the chemotaxis network modulates flagellar rotational bias in *Helicobacter pylori*. *eLife*. doi:[10.7554/eLife.63936](https://doi.org/10.7554/eLife.63936)
- Katiyar A*, [Antani JD](#)*, McKee BP, Gupta R, Lele PP, Lele TP (2021) A method for direct imaging of x-z cross-sections of fluorescent samples. *Journal of Microscopy*. doi:[10.1111/jmi.12965](https://doi.org/10.1111/jmi.12965) (*equal contribution)
- Wong GCL, [Antani JD](#), et al. (2021) Roadmap on emerging concepts in the physical biology of bacterial biofilms: from surface sensing to community formation. *Physical Biology*. doi:[10.1088/1478-3975/abdc0e](https://doi.org/10.1088/1478-3975/abdc0e)
- Ford KM, [Antani JD](#), Nagarajan A, Johnson MM, Lele PP (2018) Switching and torque generation in swarming cells of *E. coli*. *Frontiers in Microbiology*, 9, 2197. doi:[10.3389/fmicb.2018.02197](https://doi.org/10.3389/fmicb.2018.02197)
- [Antani JD](#), Gupta R, Lele PP. The chemotaxis conundrum for a polar-flagellated bacterium. *In preparation*.
- Katiyar A, Zhang J, [Antani JD](#), Yu Y, Lele PP, Reinhart-King CA, Sniadecki NJ, Roux KJ, Dickinson RB, Lele TP (2022) Lamin A/C mediated invaginations in the nuclear surface allow the nucleus to pass unimpeded through a dense array of fiber-like obstacles. *Advanced Science*, Accepted. Preprint doi: [10.1101/2022.03.10.483838](https://doi.org/10.1101/2022.03.10.483838)

Peer Review (Independent)

Dec 2021	<i>European Biophysics Journal (EBOJ)</i>
May 2022	<i>Physical Biology</i>

Selected Conference Presentations

Antani JD, Emonet T, Turner P. Dynamic target search by bacterial viruses on host surfaces	
Jun 2022	<i>Microbe</i> - Annual meeting of the American Society for Microbiology, Washington, D.C. (US)
May 2022	Yale Postdoctoral Association Symposium, Yale University, New Haven (US)
Antani JD, Lele PP. Modulation of chemotactic signaling in bacteria by mechanical forces.	
Mar 2019	ChEGSA Annual Symposium, Texas A&M University, College Station (US)
Oct 2018	American Institute of Chemical Engineers (AIChE) Annual Meeting, Pittsburgh (US)

Antani JD, Lele PP. Asymmetric Swimming and Chemotaxis in *Helicobacter pylori*.

Jan 2021 | Bacterial Locomotion And Signal Transduction (BLAST) XVI Meeting, Zoom (online)

Feb 2020 | Biophysical Society (BPS) Annual Meeting, San Diego (US).

Jan 2019 | Bacterial Locomotion And Signal Transduction (BLAST) XV Meeting, New Orleans (US)

Antani JD, Venkatesh KV. Analysis of Glucose Homeostasis in Humans.

Nov 2015 | Futuristic Approach for Alternatives to Animal Testing [CEFIPRA](#) seminar, IIT Bombay, India

Honors & Awards

2020	CIRTL Associate Certificate, Center for the Integration of Research, Teaching, and Learning
2015	Best Poster Award, AMAT seminar, IIT Bombay
2015	Undergraduate Research Award URA01, IIT Bombay
2012	All India Rank 1427 in Joint Entrance Examination for engineering, among 479,651 participants
2008	National Talent Search Scholarship, NCERT, India

Technical Skills

Perfusion & Thermal Control Experiments	Prototyping & 3D-printing	Microscopy: Dark-field, TIRF, Optical
Bacteria & Phage Culturing, Cloning	Microfluidics Experiments	Tweezers, Image Analysis

Computer Proficiency / Computational Skills

Image Processing & Analysis:	MATLAB, Python, ImageJ – {Highlight: particle-tracking algorithms}
Genetic Analysis:	Molecular Biology {SnapGene, Benchling}, Whole Genome Analysis { <i>breseq</i> }
Design & Development:	SolidWorks, AutoCAD, LabVIEW,
Simulation Packages:	ANSYS Fluent, Aspen Plus, OpenFOAM, COMSOL, CHARMM, VMD
Programming:	MATLAB, Python, C, C++, Fortran, HTML

Current and Doctoral Research

May 2021- Now	Postdoctoral Research: Host-virus Interaction Biophysics in Bacteria <i>Evolution of target search by bacteriophages on host bacterial surfaces</i> - Established a protocol for labeling phages (the viruses of bacteria) with a non-specific fluorescent dye - Visualizing labeled phage (bacterial virus) particles attaching to host cells, through epifluorescence and HILO microscopy - Set up MATLAB-based statistical analysis of single phage trajectories to quantify attachment dynamics - Employing Bayesian inference to estimate the diffusive states of phage particles from single trajectories - Comparing the attachment dynamics of ancestral and coevolved phage-host pairs <i>Evolution of resistance in E. coli against a flagellotropic phage</i> - Evolved mutants resistant against phage χ, a virus that uses rotating bacterial flagella as receptors - Trained an undergraduate to carry out the experimental evolution protocol, mentored him in writing a successful fellowship application about this project - Characterizing the genotype and phenotype of the evolved mutants to determine the evolutionary pathway(s) <i>Development of phage therapy against gastric cancer agent Helicobacter pylori</i> - Devising a method to isolate novel phages that infect <i>H. pylori</i>, a human pathogen - Wrote+writing several fellowship and grant applications to fund the original research projects
Aug 2016- May 2021	Doctoral Thesis: Sensory Functions of the Bacterial Flagellar Motor <i>Molecular mechanisms of host substrate sensing by bacterial flagellar motors</i> - Determined the effect of mechanical forces on chemotactic output of the bacterial flagellar motor, with phase microscopy, optical tweezers, and TIRF imaging - Developed a biophysical model for the regulation of allosteric interactions by mechanical force <i>Motility, chemotaxis, and random spread of Helicobacter pylori, the carcinogenic pathogen</i> - Devised a microscopy-based technique to experimentally determine the chemotactic activity of <i>H. pylori</i> - Studied the biophysical principles governing chemotactic migration by this bacterial species which employs run-reversal motility – different from the traditionally studied <i>E. coli</i>

Design and development of experimental and analytical framework

- Designed 3D-printed imaging chambers for specific requirements in microscopy experiments, e.g., perfusion, imaging side-views
- Wrote MATLAB-based image analysis codes for tracking motile bacteria, mammalian cell-nuclei, and fluorophores

Additional Research Experience

Jul 2015- Feb 2016	Machine Learning Classification of Glucose Homeostasis in Diabetic and Healthy Individuals <i>Venkatesh Lab, Chemical Engineering, IIT Bombay</i> • Formulated a Hill function-based empirical biochemical model of body glucose concentration following a fixed food-intake, obtained least-squares fits to data acquired from 150+ individuals • Constructed a classification tree based on fit-parameters using machine learning in MATLAB; established a statistical grouping among Diabetic and Healthy individuals
May-Jul 2015	A Novel Protocol for Coating Lipid Bilayers onto Silver Nanocubes <i>Hung-Jen Wu Lab, Texas A&M University</i> • Established a tip-sonication protocol to make small unilamellar vesicles of lipids • Coated the unilamellar vesicles as supported lipid bilayers onto Silver nanocubes • Employed a Localized Surface Plasmon Resonance-based nanocube sensor to monitor protein binding on lipid bilayers • Analyzed binding kinetics of GM1 ganglioside with Cholera Toxin Subunit B
May 2014- Apr 2015	Simulation-Design of a Chemical Vapor Deposition Reactor <i>Mahesh Tirumkudulu Lab, IIT Bombay</i> • Simulated fluid flow around a disk, successfully obtained the pressure coefficients observed experimentally in literature. • Proposed an unconventional design for uniform coating of parylene polymer onto silicon micro-wafers in a micro-scale Chemical Vapor Deposition reactor. • Modeled Parylene flow using ANSYS Fluent, enhanced the proposed design.
Jun 2013	Process Improvement and Technology in Jaggery-Making <i>Sanjay Mahajani Lab, IIT Bombay</i> • Examined units under development for continuous production as well as recuperation in manufacturing Jaggery, the traditional non-centrifugal cane sugar. Analyzed Jaggery samples for acidity, moisture content, water activity, density, sugar content etc; observed the XRD and ESEM techniques.
Key Course Projects	Determining Diffusion Coefficient of Microparticles using Particle-Tracking Algorithms <i>Course: Colloidal Applications and Biomolecular Interactions in Life Science (Spring 2018)</i> • Employed particle-tracking algorithms in MATLAB to track the Brownian diffusion of submicron-sized particles recorded using phase-contrast microscopy • Determined the diffusion coefficient of particles by calculating the average mean-squared displacement of an ensemble Proposal: Evolution of Bacterial Flagellum <i>Course: Evolutionary Bioinformatics (Spring 2017)</i> • Proposed a study involving the comparison of flagellar genome across multiple bacterial species which has potential to aid in obstruction of pathogenic motility • Carried out BLAST alignment analysis for nucleotide and protein sequences of flagellin (flagellar filament) gene as preliminary results Review: How Deep a Cell Can Sense <i>Course: Cell Mechanics (Spring 2016)</i> • Carried out a literature survey and reported theoretical, in-vitro, and in-silico studies on quantified investigations about the range of mechanical signaling On-line Estimation of Concentrations in a Bio-Reactor <i>Course: State Estimation (Fall 2015)</i> • Generated true states from a model based on substrate uptake, phosphorus repression and effect of seed culture on glyco-peptide antibiotic production • Implemented the Simple, Extended, and Unscented versions of the Kalman Filter to estimate the concentrations

Teaching & Mentoring Experience

Fall 2019	A hands-on workshop for image processing and single particle tracking in MATLAB <i>Pushkar Lele Lab, Texas A&M University</i>
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Undergraduate students mentored

- Oct 2021-
Now | [Austin Theroux](#), Ecology & Evolutionary Biology, Yale University
Studying the evolution of motile bacteria against a bacteriophage that attaches through the flagellum
Presenting a poster about this work at *Microbe* ASM meeting at Washington DC in June, 2022
- Dec 2019-
Jan 2021 | [Anita Sumali](#), Biomedical Engineering, Texas A&M University
MATLAB-based quantitative analysis of the run-reversal motility of *H. pylori*; co-author on [Antani et al., eLife, 2021](#)
- Apr 2017-
Apr 2018 | AggieE_Challenge – Undergraduate Mentoring, Texas A&M University
Graduate-mentor to a multidisciplinary team of eight undergraduate students employing 3D-printing technology to design scaled-up bacterial models, in order to study the hydrodynamic effects of cell morphology on motility

Teaching Assistant at Texas A&M Chemical Engineering

- Spring 2019 | Heat Transfer Operations
Spring 2017 | Bioprocess Engineering
Fall 2016 | Bioprocess Engineering

Teaching, Mentoring, & Management Training

- Sep 2021-
Dec 2021 | Auburn University's Preparing Future Faculty Workshop
Interactive workshops consisting panel discussions from faculty and university administrators
- Sep 2021 | [Fundamentals of Inclusive Teaching](#)
[Poorvu Center: Graduate and Postdoctoral Teaching Development, Yale University](#)
A 4-part workshop on creating a respectful, supportive, equitable, and transparent classroom environment
- Aug 2018-
Dec 2019 | [Academy for Future Faculty \(AFF\)](#), Center for Teaching Excellence, Texas A&M University
Faculty Mentor: Dr. Micah Green, Texas A&M University
A program designed for training graduate students as future faculty members, featuring weekly seminars and workshops, and anchored by faculty mentorship
- Jul-
Aug 2019 | An Introduction to Evidence-Based Undergraduate STEM Teaching
Multi-university Massive Open Online Course from the NSF-founded CIRTl network
Eight-week online course on [edX.org](#) catering introduction to effective teaching strategies and the research that supports them. Attended local learning community meetings for the course at Texas A&M University.
- Oct 2019 | Improving Research Mentoring Relationships – Introduction to Effective Mentorship
Center for the Improvement of Mentored Experiences in Research (CIMER)
1-day workshop at Texas A&M University, conducted by CIMER from University of Wisconsin-Madison

Leadership & Management Experience

- Jan 2022-
Now | Communications Manager, [Yale Postdoctoral Association](#)
 - Managing website profiles and social media for the YPA
 - Linking outside organizations to the YPA community
- Jun 2021 | [People Management Skills](#) – Course, Chartered Institute of Personnel and Development, UK
- Oct 2018-
Aug 2020 | [Secretary \(2018-19\) & Chair \(2019-20\), Chemical Engineering Graduate Student Advisory Council](#)
Artie McFerrin Department of Chemical Engineering, Texas A&M University
 - Designed the council by-laws as a founding member
 - Set up channels of communication through surveys and anonymous feedback mechanisms, between grad students and administration in the department
 - Addressed grad student welfare issues including salary, benefits, mental well-being, work-life balance, student-advisor relationships, facilities, career events, and social events
- Apr 2014- | Competitions Manager, AZeotropy 2015

- Mar 2015 | *IIT Bombay's Intercollegiate Chemical Engineering Symposium*
- Pioneered a new model-making competition *ChemETimer* based on time-precision
 - Designed a presentation-based competition *Swaasthya* on engineering-applications to healthcare
 - Designed *Cipher*, the online cryptic hunt which received a participation of 400+
 - Actualized unconventional chemical engineering quizzing for *Chem-O-Philia*, national level quiz

Science Outreach Volunteering Experience

- Feb 2022-
Now | [Associate Writer & Editor, LabLeaks](#)
A biweekly mailing list summarizing recent noteworthy scientific articles to the general audience.
- Feb 2022 | [Exploring Science for Middle School Students](#)
Presented research topic to middle school children in the New Haven County, Connecticut
- Jul 2020 | [Coronasurveys: Monitoring the Incidence of COVID-19 via Open Surveys](#)
Volunteered services as the primary translator of the survey to the Gujarati language.